Sensors Design Decision Report

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LHN

Ruby Fusca, Grady Emerson

Version Control

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1. Overview

The choice of sensors for the Local Hazard Network is crucial to the entire system. It informs what data is collected, the shape of the detection modules, and the power requirements. The sensors also determine the placement of the modules, as the types of sensors vary in their data-collection modes. This report will investigate each hazard, the indicators associated with it, and how those indicators can be measured. It will also discuss the relevance of each hazard with respect to the project's study areas, outlined in Figure 1.



Figure 1. Project Study Locations

The hazards we are aiming to detect with the local hazard network are flooding, bushfires, landslides, severe storms, and droughts. Each hazard has a variety of indicators that can be measured to help predict the likelihood of an extreme weather event occurring. Once these are identified, different sensor options were ranked against the design requirements and criteria to choose the most appropriate option. A summary of these results can be found in section 4 of this report.

2. Hazard Investigations

2.1 Bushfires

2.1.2 What is the hazard? How does it affect the areas we are looking at?

Bushfires are classified as unplanned, uncontrolled fires burning in vegetation. For this study, bushfires will be the umbrella term used to refer to both grassfires and bushfires. Bushfires are prevalent in Australia because of the climate and the high flammability of native vegetation, such as eucalyptus. Climate change is causing these conditions to become more extreme, leading to a higher bushfire risk across the world. In the Northern Rivers and Mandagery (study focus areas), there is not a high bushfire risk, but it has been chosen as a focus of this study because of its larger impact across other rural areas of Australia.

Figure 2 shows the severity of the black summer fires during the 2019-2020 season in the north coast and tablelands of NSW. The northern rivers study area experienced a small area of low to high severity. The overall extent of the fires across this region was significant and had a substantial impact on the environment, economy, and communities within these regions. The afac seasonal bushfire outlook highlighted the Mandagery region (south of Dubbo) as having an increased risk of fire for the 2025-2026 season, as seen in Figure 3.

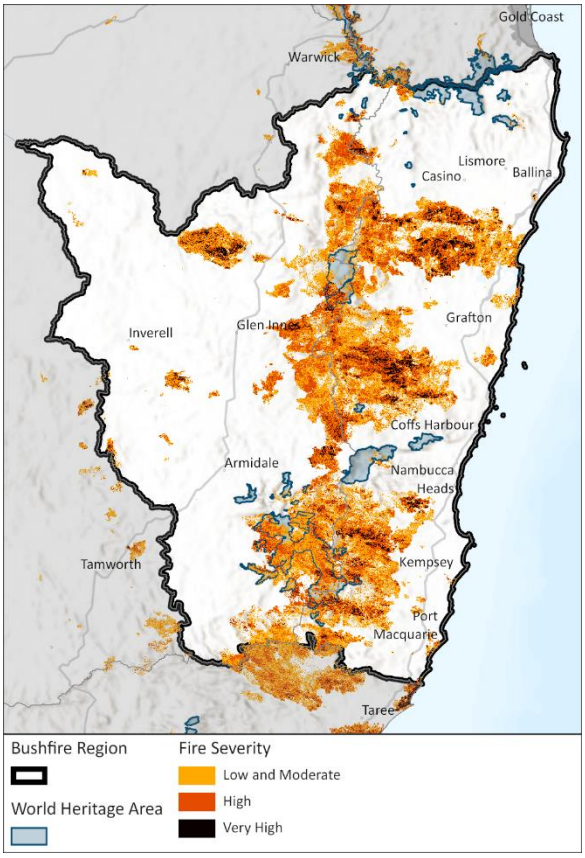


Figure 2. 2019-2020 NSW north coast and tablelands fire extent and severity map (Department of Climate Change, Energy, the Environment and Water, 2026)

2.1.3 Indicators

There is a higher risk of bushfires during hot, dry, windy weather. This is because it dries the vegetation out, making it more flammable. When the moisture in bushfire fuel is less than 20% (moisture by weight) it is easily ignited, in comparison to 30% moisture, where it is almost impossible to ignite (Sullivan, 2015). The peak bushfire conditions can occur at different times of the year across Australia with the peak danger seasons are outlined in Figure 4. Monitoring the conditions that increase the likelihood of bushfires allows for more informed decisions regarding risk management and preparation. Indicators that can increase the potential of a fire starting are temperature, relative humidity, wind speed and wind direction. Wind speeds over 12 to 15 km/h cause a significant increase in bushfire behaviour and movement (Geoscience

Australia, 2025). After a fire has started, it can be sensed through gas and particulate matter (PM) monitors due to the smoke that is produced. Often, many fires are started by a lightning strike. Lightning can be predicted using ground-based antennas networks to detect radio waves (National Oceanic and Atmospheric Administration, 2015). Lightning detection is out of scope for the project due to the number of sensors needed in the network to accurately predict lightning strikes occurring.

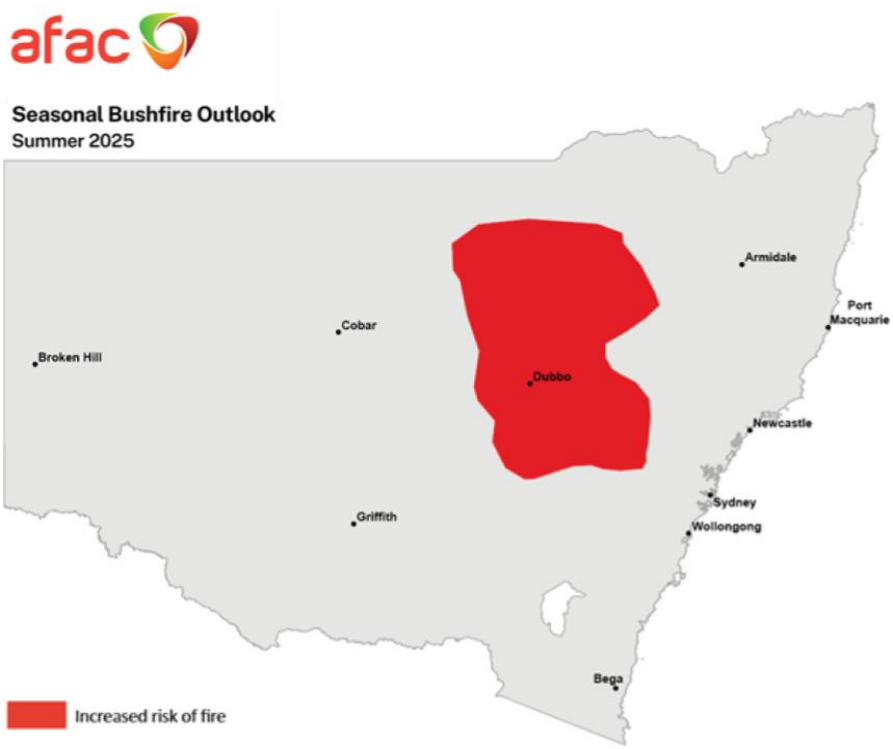


Figure 3. NSW Seasonal Bushfire Outlook Summer 2025 (Australian and New Zealand Council for fire and emergency services, 2025)

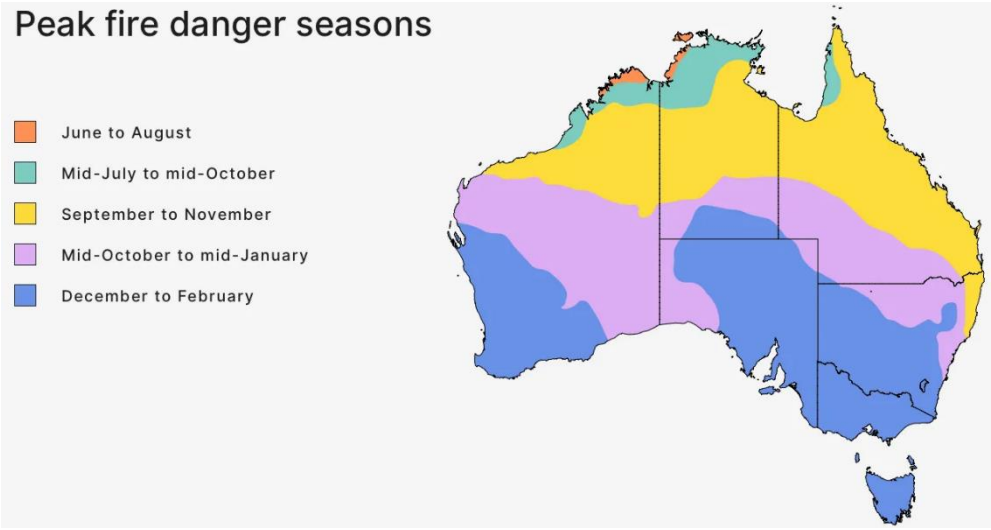


Figure 4. Peak fire danger seasons across Australia (The Bureau of Meteorology, 2026)

Currently, many research groups and companies are researching how to better predict and faster detect active fires. The Australian Institute for Disaster Resilience is looking

to develop an integrated system with 6 programs to protect Australia from bushfires. The programs include the following:

- prevention and risk reduction using satellite data to monitor eucalypt fuel conditions,
- lightning forecasting and detection,
- ignition detection using ground-based low-power IoT (Internet of Things) enabled technologies (more detail on this section specifically can be found here) and automated detection using AI and cameras on towers, and drone networks,
- a data analytics and communications platform,
- safe and rapid suppression through a GPS-guided water glider, and
- a legal and economic analysis (Australian Institue for Disaster Resilience, 2021)

For our project we are looking into developing a ground-based low-power sensor network that can detect and improve situational awareness around bushfires and other hazards.

Companies such as Attentis and Dryad have developed sensors that are designed for ultra-early detection by sensing carbon monoxide and other gases emitted during the smouldering stage (very beginning) of a bushfire (Attentis Pty Ltd, 2025) (Dryad Networks GmbH, 2026). The Dryad Silvanet Wildfire Sensor uses the Bosch BME688 gas sensor to measure CO, CO_2, H_2, VOCs, temperature, humidity, and air pressure (Dryad, n.d.).

2.1.4 Sensors

The main indicators that were found during the study and are measurable through a ground-based sensor network are listed below:

- Air temperature
- Relative humidity
- Wind speed
- Wind direction
- Soil moisture
- Gas detection (CO2, CO, trace amounts of SO2 and NO2)
- PM (particulate matter) monitoring (bushfires typically produce PM2.5L)

The optimal place to put bushfire sensors are in areas of high risk that are known to have low soil moisture. Low soil moisture indicates an increase in potential fire fuel and leads to easier ignition. Fires spread faster uphill, and in combination with the fact that smoke rises, areas of higher elevation would also be beneficial for detection.

2.2 Droughts

2.2.2 What is the hazard? How does it affect the areas we are looking at?

Droughts in Australia are considered a prolonged, abnormally dry period. i.e. when there is insufficient water for normal use. Usually, a drought is measured over at least a 3-month period, but this can vary. A flash drought is a drought that lasts a few weeks, a meteorological drought lasts for months, and a hydrological drought lasts several years. Droughts occur when there is a lack of rain, leading to a decrease in total water storage across the environment. The start of a drought begins when the expected rainfall for that region does not occur, causing soil to dry out. Following this, temporary bodies of water, such as ponds, wetlands, and ephemeral creeks, shrink or empty. For longer periods of drought, groundwater levels also drop and take a long time to recharge (eWater CRC, 2025).

The Northern Rivers area and Mandagery have both experienced droughts previously, including the Tinderbox drought from 2017 – 2019 (Murray-Darling Basin Authority, 2025). Currently, both regions have slightly decreased amounts of rainfall compared to historical data over the same period. This could lead to a drought period if conditions persist. Figure 5 shows the root zone soil moisture as of the 27th April 2026, with below average rainfall seen across the state.

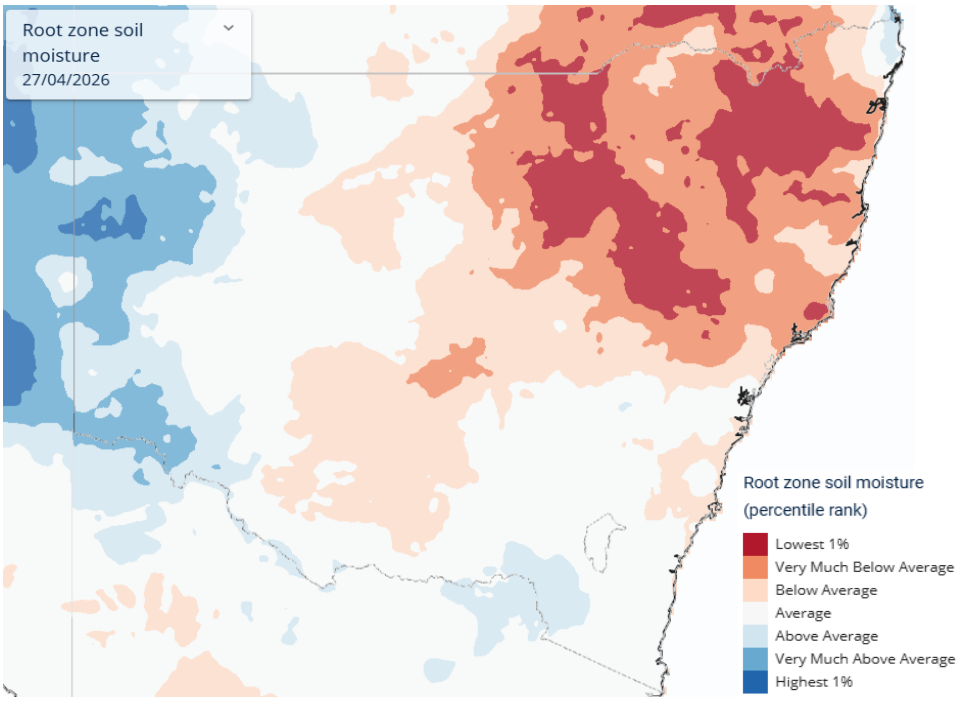


Figure 5. Root Zone Soil Moisture for NSW 27/04/2026 (Bureau of Meterology, 2026)

Figure 6 shows the combined drought indicators for each of the project's focus areas. The areas near Mandagery (Eugowra and Parkes) are both currently drought-affected with intensifying conditions, the north coast region is mostly unaffected, and a small region near the northern rivers is also showing intensifying drought conditions. The conditions in the North Coast region will likely decline further according to the

Department of Primary Industries and Regional Development (DPIRD) drought forecast (NSW Government, 2026). The combined drought indicator uses rainfall, soil moisture, plant growth, and drought direction to indicate the different phases of drought.

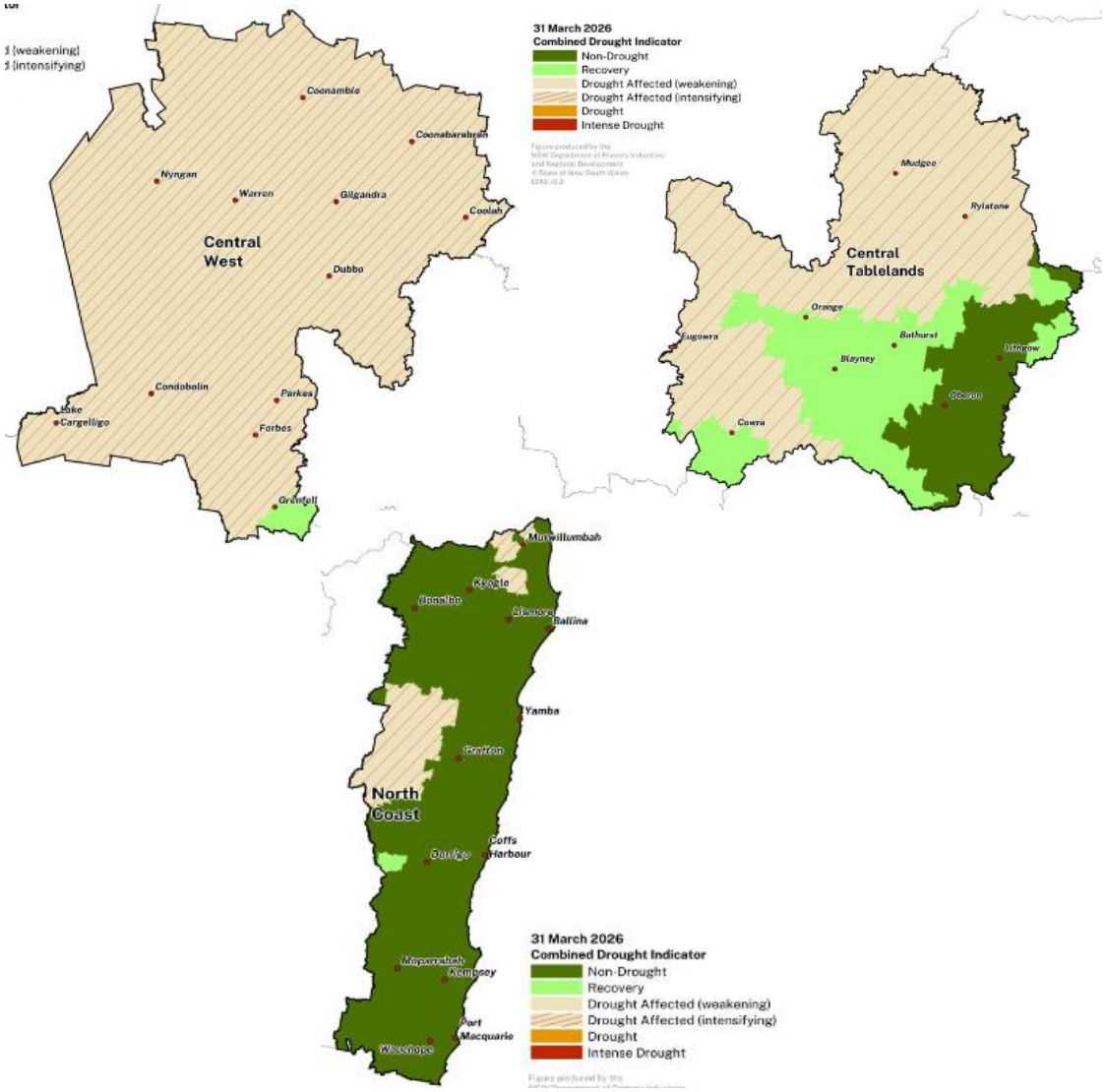


Figure 6. Drought Indicator Maps as of the 31st March 2026 for Project Focus Areas (NSW Government, 2026)

2.2.3 Indicators

Climate factors that can affect the impact and severity of a drought include the El Nino – Southern Oscillation, the positive phase of the Indian Ocean Dipole, and climate change. These conditions all impact the weather patterns within Australia and the amount of rainfall that occurs each season. Indicators that a drought is likely to occur in an area, other than decreased rainfall, include soil moisture and vegetation moisture. These are currently mainly monitored through satellite imagery (Wright, 2019).

2.2.4 Sensors

Sensors that can be used to detect drought indicators are rainfall and soil moisture. Soil moisture sensors can come in many forms, including tensiometers, watermark granular matrices, and capacitance sensors.

The depth of the soil moisture sensor affects the values of the data produced. Shallower depths (0 cm - 20 cm) show more variability with rainfall and temperature. Deeper in the soil (20 cm – 100 cm) aligns with the root zone of many plants and is a better gauge of when moisture in the soil is lacking and drought may be likely (National Integrated Drought Information System, 2024) (Agriculture Victoria, 2017).

Rainfall sensors should be placed at a distance of at least twice the height of any obstruction, preferably 4 times the height, as outlined in Figure 7.

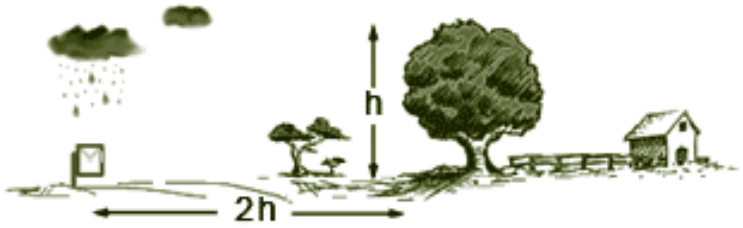


Figure 7. Rain gauge location requirements (Bureau of Meteorology, 2010)

This reduces potential error caused by nearby buildings, trees, or fences. The sensors should also be placed low to the ground (< 1 m) to reduce any wind interference (Bureau of Meteorology, 2010).

2.3 Floods

2.3.2 What is the hazard? How does it affect the areas we are looking at?

The Australian Government's standard definition of a flood is:

"The covering of normally dry land by water that has escaped or been released from the normal confines of: any lake, or any river, creek or other natural watercourse, whether or not altered or modified; or any reservoir, canal, or dam" (Geoscience Australia, 2025)

According to the Bureau of Meteorology, floods are defined to be minor, moderate or major (The Bureau of Meteorology, n.d.). The Bureau's flood warnings are informed by river heights, with a river height being the height observed on a river height measuring gauge. Observed river heights are based upon a local reference point, i.e. a typical river height measurement (The Bureau of Meteorology, n.d.).

Terania Creek, near Lismore, is located close to Leycester Creek and Wilsons River. Lismore is also within the Richmond River catchment area (NSW State Emergency Service, n.d.b), which can contribute to flooding in the region.

Main Arm is a part of the Byron Shire, with the flooding in the Byron region occurring due to the Brunswick River, Mullumbimby Creek and the Brunswick Estuary (NSW State Emergency Service, n.d.a).

Uki is a part of the Tweed Shire, where rainfall flows from the Wollumbin/Mount Warning caldera mainly into the Oxley, Rous and Tweed Rivers which can cause flooding in the area (NSW State Emergency Service, n.d.c).

As for Mandagery, it is a part of the Parkes Shire Council (Parkes Shire Council, n.d.), this area sits within the Macquarie-Bogan Rivers catchment. However, flood studies have not been yet conducted (SES, 2026).

2.3.3 Indicators

Although La Nina generally brings more water to north-eastern Australia and with this in combination with climate change predicted to increase the intensity of rainfall, this does not necessarily translate to floods occurring in rural areas and large river basins. In these circumstances, rainfall over multiple days and the soil moisture levels prior to the rainfall contribute to potential floods (CSRIO, 2024).

As for the time of year, Australia's Higher-Risk Weather Season is between October and April, which includes several natural disasters such as bushfires, heatwaves and tropical cyclones (National Emergency Management Agency, 2025).

As discussed earlier, an observed river height is a piece of data that affects flood warnings, this was considered as a potential method of detecting a potential flood. However, this would limit the positioning of the module to be in proximity to a river,

significantly reducing the areas that can be monitored. Hence, this method of detecting will not be utilised.

Measuring soil moisture content would be a valid alternative to using river height data. Lismore is considered a combination of rural and urban area, Main Arm is part of the "Rural North West" (.id community, n.d.b), Uki having a population of 685 (Australian Bureau of Statistics n.d.b) is defined as Rural by the Australian Statistical Geography Standard (ASGS) Edition 3 (Australian Bureau of Statistics 2021), and finally, as Mandagery has a population of 82 (Australian Bureau of Statistics, n.d.a) the area is also considered rural. Therefore, soil moisture levels will be considered as the indicator to be monitored for floods.

2.3.4 Sensors

As discussed in the section above this project will use soil moisture sensors to detect flood risk due to prevent location limitations. Multiple soil depths should be measured to determine at what level the soil is saturated, and a flood is likely to occur. Further details about the soil moisture sensors are outlined in section 2.2.4.

To accurately detect soil moisture in relation to floods the sensors should be in upstream areas that indicate how much water is moving through the landscape towards the catchment outlet. Additionally, low-lying areas of areas of depression are the most likely to reach saturation first and should be monitored accordingly. Soil moisture of areas alongside a river and creek (floodplains) also indicates the level of the water table and the likelihood of flooding.

2.4 Landslides

2.4.2 What is the hazard? How does it affect the areas we are looking at?

By the Queensland Government, landslides are described as being “the movement of large amounts of earth, rock, sand or mud, or any combination of these” (Queensland Government, n.d.).

According to the NSW Government, for the areas of focus, there was no apparent landslide risks (NSW Government, n.d.b). However, it is more likely that there is simply just a lack of data for said areas. For this project the assumption will be made that the areas of focus are at risk of landslides, as they were identified as a hazard of interest. This thinking can be backed by the nature of what makes an area have a landslide risk being, which commonly in NSW is caused when rainfall causes soil to saturate beyond what the vegetation can withstand the added weight, causing a landslide (Lismore City Council, n.d.). Meaning that what defines an area having a landslide risk is quite dynamic, however, as will be elaborated in the next section, can still be monitored for.

2.4.3 Indicators

As mentioned in the previous section, as excessive rain can cause landslides, looking at the time of year that has the most rainfall would be a beneficial indicator. In Lismore, the months where the average rainfall amount is above 111.93mm (yearly average divide by twelve) is between December and May (Bureau of Meteorology, 2026), which covers the rainfall of Terania. The closest weather station near Main Arm, being Fairview Farm in Mullumbimby, records that the only months that have above the yearly average divided by twelve (148.93mm) is December to May as well (Bureau of Meteorology, 2026). For Uki, the months where the average rainfall was above the yearly average divided by twelve (128.02mm) was December to March (Bureau of Meteorology, 2026). Lastly, for Mandagery, the average annual rainfall divided by twelve was 59.11mm, and the months where the averages were higher than that value was February, March, June, July and November (Bureau of Meteorology, 2026).

Soil moisture is also an indicator that a landslides may occur, therefore, recording of soil moisture levels would be a useful indicator to keep track of when considering landslide detection.

Many landslides are also caused by earthquakes. Where earthquakes shake the ground weakening it over time, allowing more rainfall to saturate the ground increasing the weight. However, large earthquakes are not common across the focus areas of interest, therefore, this will not be considered as an influencing factor in landslide detection.

2.4.4 Sensors

Much like the detection of floods, such as soil moisture and rainfall sensing has already been discussed in section 2.3.4. In addition to soil moisture and rainfall, tilt or vibration

sensors could be used to detect landslides occurring in remote areas where people are not present via an inertial measurement unit, or tilt sensor.

The location for a landslide prediction/detection sensor should be along a slope of potential risk. The soil moisture should be measured for the top points of the slope for different depths. Tilt sensors can be placed at different points along the top and side of said slope to detect if the ground is moving and angling down. Seismometers could also be used to detect ground vibrations from an active landslide occurring (USGS, 2026).

2.5 Severe Storms (Wind and/or Hail)

2.5.2 What is the hazard? How does it affect the areas we are looking at?

The Bureau of Meteorology defines a severe thunderstorm as a thunderstorm meeting one of the following criteria:

- Has hailstones that are 2cm in diameter or larger
- Has wind gusts that are 90km/h or faster (called damaging wind gusts)
- Causes a tornado
- Causes heavy rain that could cause flash flooding

If a severe thunderstorm crosses the following thresholds of having any of the following:

- Hailstones that are 5cm in diameter or larger
- Has wind gusts that are 125km/h or faster (called destructive wind gusts)
- Causes a tornado
- Causes rain that is heavy enough to cause life-threatening flash flooding

It is considered a 'very dangerous' storm (The Bureau of Meteorology, 2026).

As severe thunderstorms are localised and can occur very rapidly, they are challenging to predict (The Bureau of Meteorology, 2026). Therefore, rather than detecting an indicator for the storm as a whole, an approach to measure various components of what makes a severe storm, "severe" and compare this against a baseline for things such as rainfall, and against the definition set by the Bureau of Meteorology for indicators that have such a threshold.

2.5.3 Indicators

Please refer to section 2.4.3 for discussion of average rainfall in the areas of focus. Another indicator for storms is a change in barometric pressure. If the barometric pressure drops suddenly, it is likely that a storm is going to occur.

2.5.4 Sensors

Rainfall and windspeed will both be used as indicators for storms along with barometric pressure. Barometric pressure can be measured using a variety of sensors. The best location for this type of sensor is a stable, dry environment that is not in direct sunlight, wind, or rapid temperature changes. This will be important to consider when developing the enclosure of the sensor nodes.

3. Requirements

Table 1 summarizes what indicators are relevant to each hazard. Rainfall and soil moisture are extremely important as they are indicators for 4+ hazards. To choose the correct sensor for each indicator, a pairwise analysis of the sensor requirements was conducted and used to highlight the most important design criteria. To see the list of all requirements and design criteria, along with their pairwise analysis, see Specifications, Requirements & Criteria.

Table 1. Indicators for each hazard

		Indicators						
		Rainfall	Temperature	Windspeed	RH	Soil Moisture	Air Quality	Barometric Pressure
Hazards	Bushfires	✓	✓	✓	✓	✓	✓	X
	Droughts	✓	X	X	X	✓	X	X
	Flooding	✓	X	X	X	✓	X	X
	Storms	✓	X	✓	✓	X	X	✓
	Landslides	✓	X	X	X	✓	X	X

Table 2 lists the requirements that were ranked as being the top 5 most important and informed the design criteria that each selected sensor had to meet. For a more detailed list of the sensor requirements see the Requirements and Criteria spreadsheet.

Table 2. Top 5 sensor requirements

Sensor Node Requirements	
srID	**Description**
SR7	Sensors must not pose an unreasonable environmental hazard or cause environmental disruption
SR8	Sensors must not pose an unreasonable safety risk
SR6	Sensors must be accurate enough to reliably detect hazards
SR3	Sensors must be able to work within the broader detection module
SR14	The sensors should make up a module that can be easily replaced

To keep the whole detection module easily deployable, unless other solutions are found, the methods of detection of the sensors should allow the whole detection system to be one unit. I.e., A particular sensor should not restrict the positioning of the whole module to the point where the other sensors are restricted/limited in their ability to record data. Another potential restriction for the detection module would be the sensor's communication protocol with the RAK module.

4. Recommendations

The sensor that ranked highest against the design criteria was the Bosch BME688. It measures pressure, humidity, temperature, and air quality. It is currently being used by DRYAD in their Silvanet Wildfire Sensor. This is an outdoor sensor used for ultra-early wildfire detection, which is a similar application to this project. Along with this sensor, the tipping bucket rain gauge, wind sensor Rev. C, and the DFRobot waterproof capacitive soil moisture sensor will be used to measure the other required indicators. Two soil moisture sensors should be used for each node to measure the moisture at two different depths, giving a better understanding of the soil profile.

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